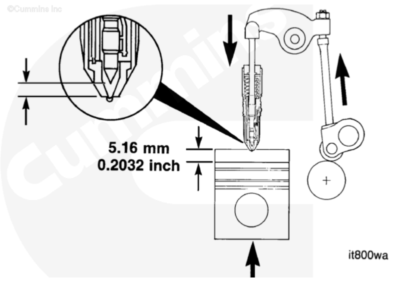
Trainings

General Information

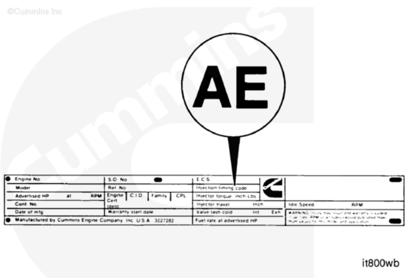
The injection timing is the relative measurement of the distance remaining between the injector plunger and the injector cup when the piston is 5.16 mm [0.2032 in], or 19 degrees before top dead center on the compression stroke.

Injector timing is expressed by the amount of push rod travel remaining.



The injector timing code appears on the engine dataplate. Codes are alphabetic letters that relate to a numerical specification.

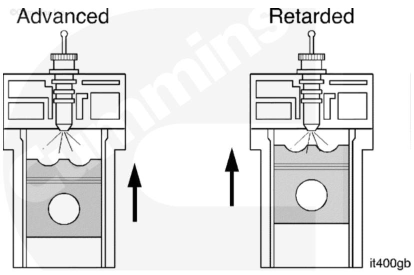
Specifications can be found in the Control Part List (CPL) Manual, Bulletin 4021328.



Below is a brief review of injection timing and how it can be adjusted.

Advanced timing means the fuel is injected earlier into the cylinder during the compression stroke.

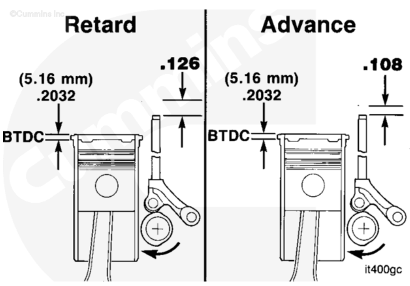
Retarded timing means the fuel injection occurs closer to top dead center in the cylinder.



The amount of push rod travel determines the time of fuel injection in relation to the piston position.

The higher the numerical value of the push rod travel remaining indicates a greater degree of retarded or slow timing.

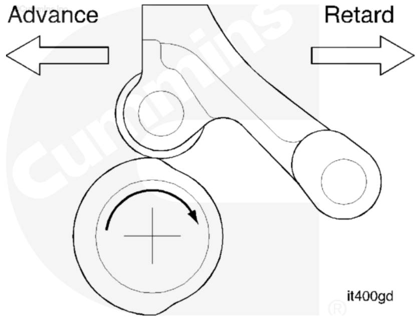
The lower the numerical value of the push rod travel remaining indicates a greater degree of advanced or fast timing.



Injection timing changes are accomplished by advancing or retarding the cam follower action in relation to the piston position.

This is accomplished by changing the orientation of the camshaft lobe to the cam follower using different camshaft gear keys.

Gear train timing (index mark alignment) always remain the same.

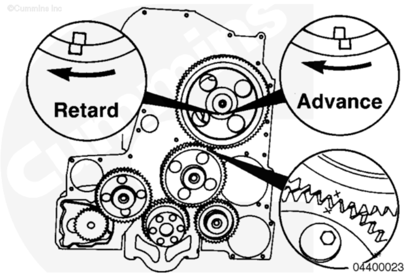


The camshaft key provides a means of indexing the camshaft with the gear.

Offset keys allow the camshaft profile to be rotated slightly while the gear train timing remains the same.

The more the top of the offset is moved in the direction of the camshaft normal rotation, the more the injection timing will be retarded. The push rod travel numerical value will increase.

The direction of normal rotation on a K19 engine crankshaft is **clockwise** as viewed from the front.



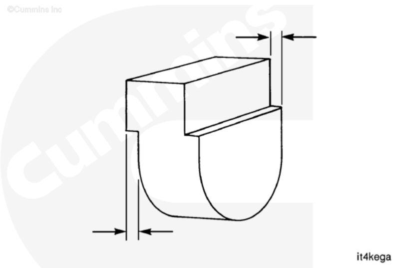
Offset keys can be identified by measuring the offset and referring to the following chart.

Each 0.025 mm [0.001 in] of offset will cause a 0.0127 mm [0.0005 in] change in push rod travel from a straight key.

If checking or setting the injection timing, it is recommended to use a testing gear. A testing gear is a camshaft gear that has been modified to provide a slip-fit on the camshaft.

The camshaft key part number listed in the engine performance parts option is a starting point. Do **not** assume the listed key will provide the specified timing. **Always** measure the injection timing if the gear, camshaft, or key have been changed or if during disassembly the direction of any offset was **not** noted. Part of the reason for the existence of the offset key is to be able to adjust the static timing to all tolerances for the part used in the engine.

If a slip fit gear is used, the injection timing **must** be measured again after installation of the production gear.

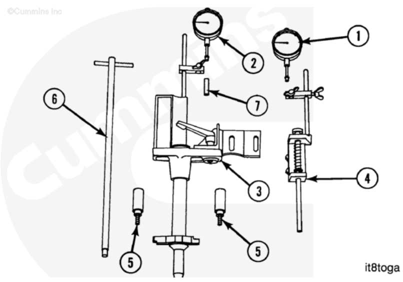


Timing Code	Recommended Code	Direction of Offset
AE	216782	Opposite camshaft rotation
AJ	200706	With camshaft rotation
AM	216782	With camshaft rotation
CI	200711	With camshaft rotation
CL	S-302	None
CU	3000492	With camshaft rotation

Measure

Use the injection timing tool, Part Number 3824942. The indicators (1) and (2) are identical.

- (1) Push rod travel indicator
- (2) Piston travel indicator
- (3) Piston plunger support assembly
- (4) Push rod plunger support
- (5) Hold-down adapter
- (6) Extension assembly (adapter wrench)
- (7) Indicator stem extension.



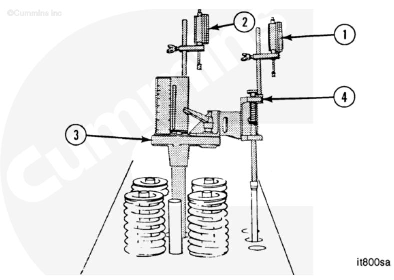
The push rod plunger support assembly alignment is critical.

Install the push rod plunger support (4) in the outside slot in the piston support (3).

Align the push rod plunger support with the mark. Tighten the capscrew.

Install indicators (1) and (2) on the posts. Turn the indicators so they are **not** over the plungers.

Install the stem extension on the piston travel indicators.

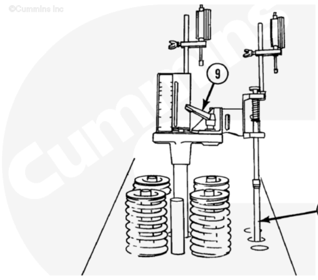


Install the injector push rod (8).

Install the timing tool in the injector bore. Install the hold-down adapter.

Align the push rod plunger and the rod to be sure they are straight.

Tighten the support lock (9).



it4toha

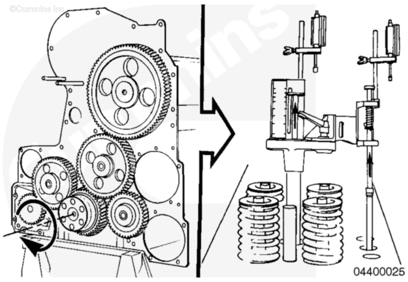
Use **only** the crankshaft to rotate the engine. The use of gears will result in false measurement. Gear lash **must** be closed up in the direction of normal rotation (crankshaft **clockwise**).

Three guide bolts equally spaced in front of the crankshaft will aid in engine rotation.

With the engine in the compression stroke, turn the crankshaft in the direction of normal rotation and observe both timing tool plungers.

Both plungers will begin moving upward when the cylinder is on the compression stroke. The indicators will be rotating in a **clockwise** direction.

If both indicators do **not** rotate in a **clockwise** direction, the engine is on the exhaust stroke. Rotate the crankshaft one revolution to get to the compression stroke.



04400025

Slowly rotate the crankshaft in the direction of normal rotation while observing the piston plunger (10). The plunger will move upward, stop, then begin to move downward. The stop point of the plunger is top dead center.

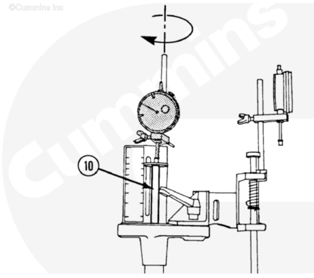
Rotate the engine opposite the direction of normal rotation until the plunger begins to move downward. The cylinder is now slightly before top dead center.

Turn the indicator so the stem is touching the plunger.

Carefully lower the indicator until it bottoms out. Raise the indicator when the needle has turned a minimum of three revolutions 7.62 mm [0.300 in]. Lock the indicator in position.

Slowly turn the crankshaft in the direction of normal rotation until the indicator needle stops turning **clockwise** (top dead center). Adjust the indicator to zero.

Always zero at top dead center with the crankshaft having just been rotated in the direction of normal rotation.

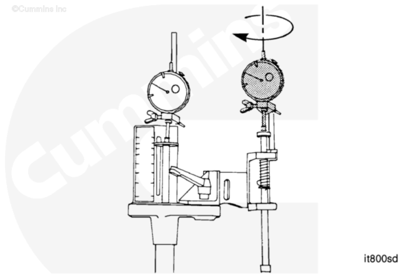


it800sc

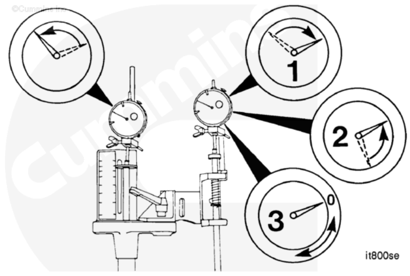
Slowly and carefully rotate the crankshaft backward and forward until the needle stops at zero before reversing the direction to indicate the piston is after top dead center.

Turn the push rod indicator so the stem touches the plunger.

Carefully lower the indicator until it bottoms out. Raise the indicator when the needle has turned a minimum of three revolutions 7.62 mm [0.300 in].



Slowly turn the crankshaft in the direction of normal rotation until the push rod indicator stops (1), momentarily reverses direction (2), (this is the crush nose on the camshaft), and stops again (3). The cam follower is now on the outer base circle of the camshaft. The piston is now approximately 90 degrees after top dead center.



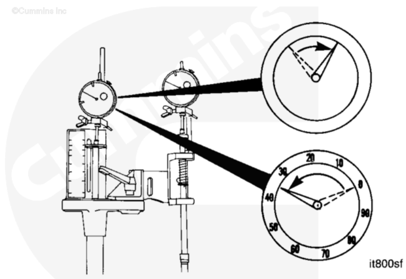
It is important to record the amount of travel remaining in the push rod travel indicator for later reference.

Carefully lower the push rod travel indicator until it bottoms out. Raise the indicator approximately ¼ revolution 6.35 mm [0.025 in]. Lock the indicator in position.

Set the indicator at zero.

Observe the piston travel indicator as the crankshaft is slowly rotated opposite the direction of normal rotation.

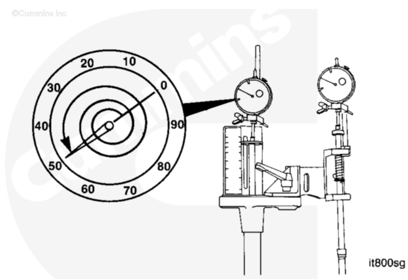
Stop rotating the crankshaft when the piston travel indicator indicates the piston is at top dead center (zero).



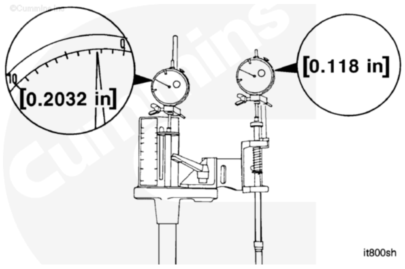
The crankshaft **must** be turned slowly to accurately count the indicator revolutions.

Turn the crankshaft opposite the direction of normal rotation until the indicator needle moves 2½ revolutions, 6.35 mm [0.250 in].

The piston is now 6.35 mm [0.250 in] before top dead center.



Only move the piston to 5.16 mm [0.2032 in] before top dead center by turning the crankshaft in the direction of normal rotation. If the crankshaft is turned too far, turn the crankshaft back opposite the direction of normal rotation more than 5.16 mm [0.2032 in] before top dead center. Then very slowly turn the crankshaft in the direction of normal rotation until the indicator indicates the piston is 5.16 mm [0.2032 in] before top dead center.



All K19 injection timing specifications are more than one indicator revolution 2.54 mm [0.100 in].

Read the push rod indicator **counterclockwise** from zero. This is the injection timing measurement. An example of 0.118 in is illustrated in the graphic.

If unsure of the number of push rod indicator revolutions, check by carefully lifting the indicator stem until the indicator has bottomed out. Lower the stem the amount of excess travel. Lower the stem to the plunger. Read the indicator.

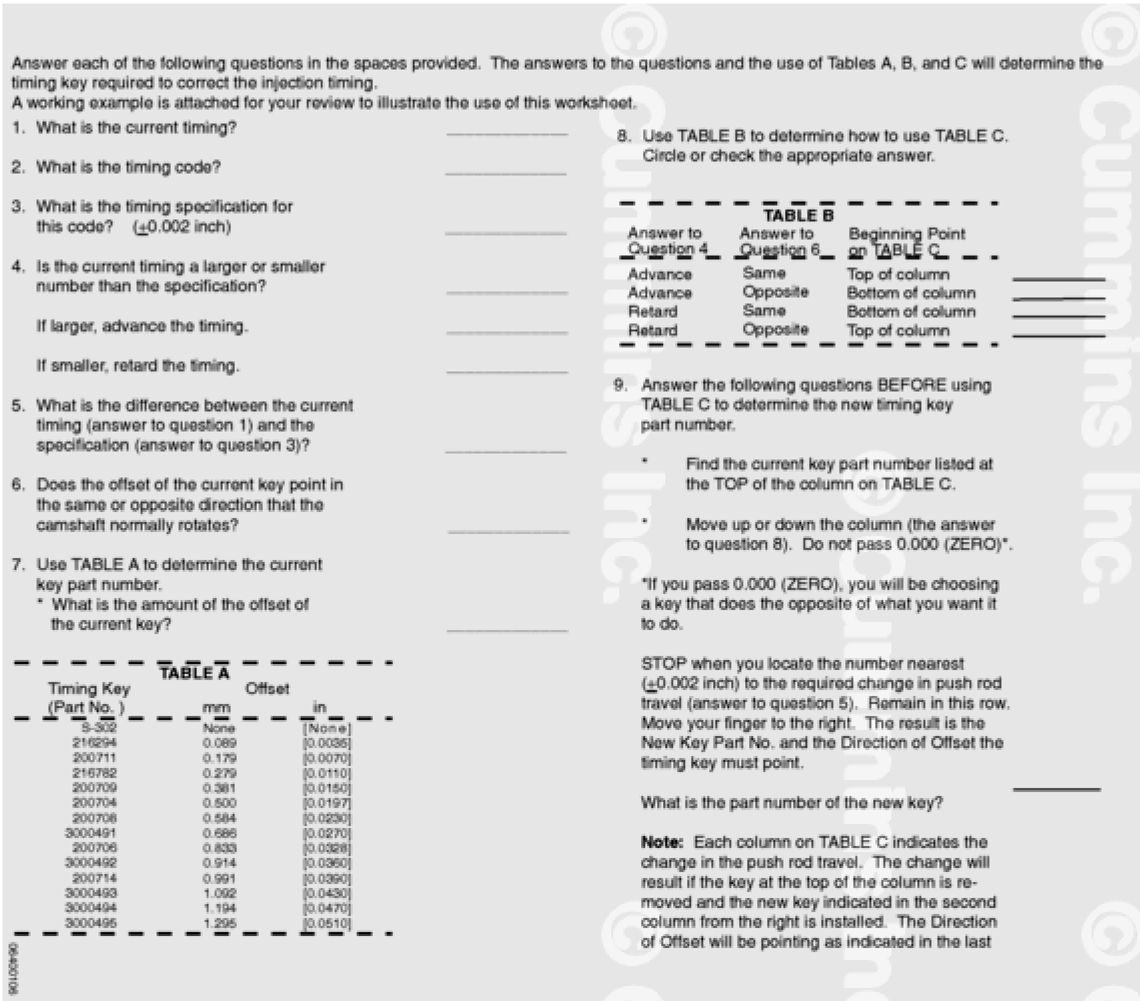
If the injection timing is within specification and a slip fit gear is used, install the standard camshaft gear. Refer to Procedure 001-012 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-012.html). Repeat the injection timing procedure after the gear has cooled.

If the injection timing is still **not** within specification, repeat the measurement procedure to check the tool set up and zero settings.

If the timing is still **not** within specification, the camshaft key **must** be changed. Remove the camshaft gear. Refer to Procedure 001-012 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-012.html).

Record the orientation of the offset of the key. Use the following worksheet to determine the alternate key.

The timing measurement **must** be confirmed after changing the key.



Answer each of the following questions in the spaces provided. The answers to the questions and the use of Tables A, B, and C will determine the timing key required to correct the injection timing.
A working example is attached for your review to illustrate the use of this worksheet.

1. What is the current timing? ____________
2. What is the timing code? ____________
3. What is the timing specification for this code? (±0.002 inch) ____________
4. Is the current timing a larger or smaller number than the specification? ____________

   If larger, advance the timing. ____________

   If smaller, retard the timing. ____________
5. What is the difference between the current timing (answer to question 1) and the specification (answer to question 3)? ____________
6. Does the offset of the current key point in the same or opposite direction that the camshaft normally rotates? ____________
7. Use TABLE A to determine the current key part number.
   * What is the amount of the offset of the current key? ____________

TABLE A

Timing Key (Part No.)	Offset mm	Offset in
S-302	None	[None]
216294	0.089	[0.0035]
200711	0.179	[0.0070]
216782	0.279	[0.0110]
200709	0.381	[0.0150]
200704	0.500	[0.0197]
200708	0.584	[0.0230]
3000491	0.686	[0.0270]
200706	0.833	[0.0328]
3000492	0.914	[0.0360]
200714	0.991	[0.0390]
3000493	1.092	[0.0430]
3000494	1.194	[0.0470]
3000495	1.295	[0.0510]

8. Use TABLE B to determine how to use TABLE C. Circle or check the appropriate answer.

TABLE B

Answer to Question 4	Answer to Question 6	Beginning Point on TABLE C	
Advance	Same	Top of column	________
Advance	Opposite	Bottom of column	________
Retard	Same	Bottom of column	________
Retard	Opposite	Top of column	________

9. Answer the following questions BEFORE using TABLE C to determine the new timing key part number.

   * Find the current key part number listed at the TOP of the column on TABLE C.

   * Move up or down the column (the answer to question 8). Do not pass 0.000 (ZERO)*.

   *If you pass 0.000 (ZERO), you will be choosing a key that does the opposite of what you want it to do.

   STOP when you locate the number nearest (±0.002 inch) to the required change in push rod travel (answer to question 5). Remain in this row. Move your finger to the right. The result is the New Key Part No. and the Direction of Offset the timing key must point.

   What is the part number of the new key? ________

   Note: Each column on TABLE C indicates the change in the push rod travel. The change will result if the key at the top of the column is removed and the new key indicated in the second column from the right is installed. The Direction of Offset will be pointing as indicated in the last

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NEEDS TEXT FOR CORRECT STRUCTURE

TABLE C

S-302	200704	200706	200708	200709	200711	200714	216294	216782	3000491	3000492	3000493	3000494	3000495	New Key	
														Part No.	Direction of Offset
0.0256	0.0351	0.0411	0.0366	0.0326	0.0289	0.0441	0.0273	0.0311	0.0391	0.0436	0.0471	0.0491	0.0512	3000495	Opposite
0.0235	0.033	0.039	0.0345	0.0305	0.0268	0.042	0.0252	0.029	0.037	0.0415	0.045	0.0471	0.0491	3000494	Opposite
0.0215	0.031	0.037	0.0325	0.0285	0.0248	0.040	0.0232	0.027	0.035	0.0395	0.043	0.0441	0.0471	3000493	Opposite
0.0185	0.028	0.034	0.0295	0.0255	0.0218	0.037	0.0202	0.024	0.032	0.0365	0.040	0.042	0.0441	200714	Opposite
0.018	0.0275	0.0335	0.029	0.025	0.0213	0.0365	0.0197	0.0235	0.0315	0.036	0.0395	0.0415	0.0436	3000492	Opposite
0.0155	0.025	0.031	0.0265	0.0225	0.0188	0.034	0.0172	0.021	0.029	0.0335	0.037	0.039	0.0411	200706	Opposite
0.0135	0.023	0.029	0.0245	0.0205	0.0168	0.032	0.0152	0.019	0.027	0.0315	0.035	0.037	0.0391	3000491	Opposite
0.011	0.0205	0.0265	0.022	0.018	0.0143	0.0295	0.0127	0.0165	0.0245	0.029	0.0325	0.0345	0.0366	200708	Opposite
0.0095	0.019	0.025	0.0205	0.0165	0.0128	0.028	0.0112	0.015	0.023	0.0275	0.031	0.033	0.0351	200704	Opposite
0.007	0.0165	0.0225	0.018	0.014	0.0103	0.0255	0.0087	0.0125	0.0205	0.025	0.0285	0.0305	0.0326	200709	Opposite
0.0055	0.015	0.021	0.0165	0.0125	0.009	0.024	0.0072	0.011	0.019	0.0235	0.027	0.029	0.0311	216782	Opposite
0.0033	0.0128	0.0188	0.0143	0.0103	0.006	0.0218	0.005	0.009	0.0168	0.0213	0.0248	0.0268	0.0289	200711	Opposite
0.0017	0.0112	0.0172	0.0127	0.0087	0.0051	0.0202	0.0034	0.0072	0.0152	0.0197	0.0232	0.0252	0.0273	216294	Opposite
0.000	0.0095	0.0155	0.011	0.007	0.003	0.0185	0.0017	0.006	0.0135	0.018	0.0205	0.0235	0.0256	S-302	N/A
0.0017	0.0078	0.0138	0.0093	0.0053	0.0016	0.0168	0.000	0.0038	0.0118	0.0163	0.0198	0.0218	0.0239	216294	Same
0.0033	0.0062	0.0122	0.0077	0.0037	0.000	0.0152	0.0016	0.002	0.0102	0.0147	0.0182	0.0202	0.0223	200711	Same
0.0055	0.004	0.010	0.0055	0.0015	0.002	0.013	0.0038	0.000	0.008	0.0125	0.016	0.018	0.0201	216782	Same
0.007	0.0025	0.0085	0.004	0.000	0.0037	0.0115	0.0053	0.0015	0.0065	0.011	0.0145	0.0165	0.0186	200709	Same
0.0095	0.000	0.006	0.0015	0.0025	0.0062	0.009	0.0078	0.004	0.004	0.0085	0.012	0.014	0.0161	200704	Same
0.011	0.0015	0.0045	0.000	0.004	0.0077	0.0075	0.0093	0.0055	0.0025	0.007	0.0105	0.0125	0.0146	200708	Same
0.0135	0.004	0.002	0.0025	0.0065	0.0102	0.005	0.0118	0.008	0.000	0.0045	0.008	0.010	0.0121	3000491	Same
0.0155	0.006	0.000	0.0045	0.0085	0.012	0.003	0.0138	0.010	0.002	0.0025	0.006	0.008	0.0101	200706	Same
0.018	0.0085	0.0025	0.007	0.011	0.0147	0.0005	0.0163	0.0125	0.0045	0.000	0.0035	0.0055	0.0076	3000492	Same
0.0185	0.009	0.003	0.0075	0.0115	0.0152	0.000	0.0168	0.013	0.005	0.0005	0.003	0.005	0.0071	200714	Same
0.0215	0.012	0.006	0.0105	0.0145	0.0182	0.003	0.0198	0.016	0.008	0.0035	0.000	0.002	0.0041	3000493	Same
0.0235	0.014	0.008	0.0125	0.0165	0.202	0.005	0.0218	0.018	0.010	0.0055	0.002	0.000	0.0021	3000494	Same
0.0256	0.0161	0.010	0.0146	0.0186	0.0223	0.0071	0.0239	0.0020	0.0121	0.0076	0.0041	0.0021	0.000	3000495	Same

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NEEDS TEXT FOR CORRECT STRUCTURE

Answer each of the following questions in the spaces provided. The answers to the questions and the use of Tables A, B, and C will determine the timing key required to correct the injection timing.
A working example is attached for your review to illustrate the use of this worksheet.

1. What is the current timing? **0.267**

2. What is the timing code? **JF**

3. What is the timing specification for this code? (±0.002 inch) **0.279**

4. Is the current timing a larger or smaller number than the specification? **Smaller**

   If larger, advance the timing.

   If smaller, retard the timing. **Retard**

5. What is the difference between the current timing (answer to question 1) and the specification (answer to question 3)? **0.012**

6. Does the offset of the current key point in the same or opposite direction that the camshaft normally rotates? **Same**

7. Use TABLE A to determine the current key part number.
   * What is the amount of the offset of the current key? **0.007**
   * What is the part number of the current key? **200711**

TABLE A

Timing Key (Part No.)	Offset mm	Offset in
S-302	None	[None]
216294	0.089	[0.0035]
200711	0.179	[0.0070]
216782	0.279	[0.0110]
200709	0.381	[0.0150]
200704	0.500	[0.0197]
200708	0.584	[0.0230]
3000491	0.686	[0.0270]
200706	0.833	[0.0328]
3000492	0.914	[0.0360]
200714	0.991	[0.0390]
3000493	1.092	[0.0430]
3000494	1.194	[0.0470]
3000495	1.295	[0.0510]

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8. Use TABLE B to determine how to use TABLE C. Circle or check the appropriate answer.

TABLE B

Answer to Question 4	Answer to Question 6	Beginning Point on TABLE C	
Advance	Same	Top of column	
Advance	Opposite	Bottom of column	
Retard	Same	Bottom of column	✓
Retard	Opposite	Top of column	

9. Answer the following questions BEFORE using TABLE C to determine the new timing key part number.

   * Find the current key part number listed at the TOP of the column on TABLE C.

   * Move up or down the column (the answer to question 8). Do not pass 0.000 (ZERO)*.

   *If you pass 0.000 (ZERO), you will be choosing a key that does the opposite of what you want it to do.

   STOP when you locate the number nearest (±0.002 inch) to the required change in push rod travel (answer to question 5). Remain in this row. Move your finger to the right. The result is the New Key Part No. and the Direction of Offset the timing key must point.

   What is the part number of the new key? **200706**

   Note: Each column on TABLE C indicates the change in the push rod travel. The change will result if the key at the top of the column is removed and the new key indicated in the second column from the right is installed. The Direction of Offset will be pointing as indicated in the last column to the right on TABLE C.

NEEDS TEXT FOR CORRECT STRUCTURE

TABLE C

S-302	200704	200706	200708	200709	200711	200714	216294	216782	3000491	3000492	3000493	3000494	3000495	New Key Part No.	New Key Direction of Offset
0.0256	0.0351	0.0411	0.0366	0.0326	0.0289	0.0441	0.0273	0.0311	0.0391	0.0436	0.0471	0.0491	0.0512	3000495	Opposite
0.0235	0.033	0.039	0.0345	0.0305	0.0268	0.042	0.0252	0.029	0.037	0.0415	0.045	0.0471	0.0491	3000494	Opposite
0.0215	0.031	0.037	0.0325	0.0285	0.0248	0.040	0.0232	0.027	0.035	0.0395	0.043	0.0441	0.0471	3000493	Opposite
0.0185	0.028	0.034	0.0295	0.0255	0.0218	0.037	0.0202	0.024	0.032	0.0365	0.040	0.042	0.0441	200714	Opposite
0.018	0.0275	0.0335	0.029	0.025	0.0213	0.0365	0.0197	0.0235	0.0315	0.036	0.0395	0.0415	0.0436	3000492	Opposite
0.0155	0.025	0.031	0.0265	0.0225	0.0188	0.034	0.0172	0.021	0.029	0.0335	0.037	0.039	0.0411	200706	Opposite
0.0135	0.023	0.029	0.0245	0.0205	0.0168	0.032	0.0152	0.019	0.027	0.0315	0.035	0.037	0.0391	3000491	Opposite
0.011	0.0205	0.0265	0.022	0.018	0.0143	0.0295	0.0127	0.0165	0.0245	0.029	0.0325	0.0345	0.0366	200708	Opposite
0.0095	0.019	0.025	0.0205	0.0165	0.0128	0.028	0.0112	0.015	0.023	0.0275	0.031	0.033	0.0351	200704	Opposite
0.007	0.0165	0.0225	0.018	0.014	0.0103	0.0255	0.0087	0.0125	0.0205	0.025	0.0285	0.0305	0.0326	200709	Opposite
0.0055	0.015	0.021	0.0165	0.0125	0.009	0.024	0.0072	0.011	0.019	0.0235	0.027	0.029	0.0311	216782	Opposite
0.0033	0.0128	0.0188	0.0143	0.0103	0.006	0.0218	0.005	0.009	0.0168	0.0213	0.0248	0.0268	0.0289	200711	Opposite
0.0017	0.0112	0.0172	0.0127	0.0087	0.0051	0.0202	0.0034	0.0072	0.0152	0.0197	0.0232	0.0252	0.0273	216294	Opposite
0.000	0.0095	0.0155	0.011	0.007	0.003	0.0185	0.0017	0.006	0.0135	0.018	0.0205	0.0235	0.0256	S-302	N/A
0.0017	0.0078	0.0138	0.0093	0.0053	0.0016	0.0168	0.000	0.0038	0.0118	0.0163	0.0198	0.0218	0.0239	216294	Same
0.0033	0.0062	0.0122	0.0077	0.0037	0.000	0.0152	0.0016	0.002	0.0102	0.0147	0.0182	0.0202	0.0223	200711	Same
0.0055	0.004	0.010	0.0055	0.0015	0.002	0.013	0.0038	0.000	0.008	0.0125	0.016	0.018	0.0201	216782	Same
0.007	0.0025	0.0085	0.004	0.000	0.0037	0.0115	0.0053	0.0015	0.0065	0.011	0.0145	0.0165	0.0186	200709	Same
0.0095	0.000	0.006	0.0015	0.0025	0.0062	0.009	0.0078	0.004	0.004	0.0085	0.012	0.014	0.0161	200704	Same
0.011	0.0015	0.0045	0.000	0.004	0.0077	0.0075	0.0093	0.0055	0.0025	0.007	0.0105	0.0125	0.0146	200708	Same
0.0135	0.004	0.002	0.0025	0.0065	0.0102	0.005	0.0118	0.008	0.000	0.0045	0.008	0.010	0.0121	3000491	Same
0.0155	0.006	0.000	0.0045	0.0085	0.012	0.003	0.0138	0.010	[illegible]	0.0025	[illegible]	0.008	[illegible]	200706	Same
0.018	0.0085	0.0025	0.007	0.011	0.0147	0.0005	0.0163	0.0125	0.0045	0.000	0.0035	0.0055	0.0076	3000492	Same
0.0185	0.009	0.003	0.0075	0.0115	0.0152	0.000	0.0168	0.013	0.005	0.0005	0.003	0.005	0.0071	200714	Same
0.0215	0.012	0.006	0.0105	0.0145	0.0182	0.003	0.0198	0.016	0.008	0.0035	0.000	0.002	0.0041	3000493	Same
0.0235	0.014	0.008	0.0125	0.0165	0.202	0.005	0.0218	0.018	0.010	0.0055	0.002	0.000	0.0021	3000494	Same
0.0256	0.0161	0.010	0.0146	0.0186	0.0223	0.0071	0.0239	0.0020	0.0121	0.0076	0.0041	0.0021	0.000	3000495	Same

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NEEDS TEXT FOR CORRECT STRUCTURE